

PROJECT REPORT

Women's Clothing: Trends and Analysis in the Fashion Industry

Task

Handling Missing Values Using KNN Imputation, MICE, and Predictive Modeling with Manual Selection of the Best Method Based on Statistical Analysis

1. Introduction

The fashion industry generates a large amount of customer and product data. Women's clothing datasets may contain information such as customer age, product ratings, recommendations, positive feedback, product categories, review titles, and review text.

However, real-world datasets often contain missing values. Missing data can reduce the quality of analysis and may affect the performance of machine-learning models.

The main objective of this project is to identify and handle missing values in the Women's Clothing dataset. Multiple techniques, including **K-Nearest Neighbors (KNN) Imputation**, **Multiple Imputation by Chained Equations (MICE)**, and **Predictive Modeling**, were evaluated. Statistical measures such as **Mean Absolute Error (MAE)** and **Root Mean Squared Error (RMSE)** were used to compare numerical imputation performance.

2. Project Objective

The main objectives of this project are:

- To load and understand the Women's Clothing dataset.
 - To identify columns containing missing values.
 - To calculate the number and percentage of missing values.
 - To classify features into numerical, categorical, and text variables.
 - To analyze the statistical characteristics of numerical variables.
 - To evaluate KNN imputation.
 - To evaluate MICE imputation.
 - To evaluate predictive modeling using Random Forest.
 - To compare the methods using MAE and RMSE.
 - To manually select an appropriate treatment according to the characteristics of each column.
 - To create a final cleaned dataset for further analysis.
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3. Dataset Description

The dataset contains information related to women's clothing products and customer reviews.

The main attributes include:

Attribute	Description
Clothing ID	Identifier of the clothing product
Age	Age of the customer
Title	Title of the customer review
Review Text	Detailed customer review
Rating	Rating given to the product
Recommended IND	Indicates whether the customer recommends the product
Positive Feedback Count	Number of positive feedback responses
Division Name	Product division
Department Name	Product department
Class Name	Product class

The dataset contains approximately **23,486 records**.

4. Missing Value Analysis

The first stage of preprocessing was to identify missing values using Python.

The following missing values were found:

Column	Missing Values	Missing Percentage
Title	3,810	16.22%
Review Text	845	3.60%
Division Name	14	0.06%
Department Name	14	0.06%
Class Name	14	0.06%

The total number of missing entries was **4,697**.

The `Title` column contained the highest proportion of missing data.

Bar charts and a missing-value heatmap were used to visualize the missing-data pattern.

5. Data Type Analysis

The attributes were divided into three major groups.

Numerical Attributes

- Clothing ID
- Age
- Rating
- Recommended IND
- Positive Feedback Count

`Clothing ID` is numeric in storage but represents an identifier rather than a continuous measurement.

Categorical Attributes

- Division Name
- Department Name
- Class Name

Text Attributes

- Title
- Review Text

This classification was important because the same imputation technique is not appropriate for every data type.

6. Statistical Analysis

Statistical analysis was performed on the numerical attributes.

Measures such as the following were calculated:

- Mean
- Median
- Standard deviation
- Minimum
- Maximum
- Skewness
- Correlation

The mean customer age was approximately **43.20 years**, while the median age was approximately **41 years**.

The mean product rating was approximately **4.20**, while its median was **5**.

Positive Feedback Count showed strong positive skewness of approximately **6.47**, indicating that most reviews received relatively low positive-feedback counts while a smaller number received very high counts.

Correlation analysis also showed a strong relationship of approximately **0.79 between Rating and Recommended IND**.

7. Controlled Imputation Experiment

The numerical variables in the original dataset did not contain missing values.

Therefore, to objectively compare KNN, MICE, and predictive modeling, a **controlled masking experiment** was performed.

Approximately **10% of known values were temporarily hidden** from selected numerical columns.

The models then estimated these hidden values.

Because the true values were already known, predicted values could be compared against the original values using MAE and RMSE.

The experiment was performed primarily on:

- Age
- Positive Feedback Count

This procedure allowed the imputation methods to be compared under the same test conditions.

8. KNN Imputation

K-Nearest Neighbors imputation estimates a missing value using observations that are similar to the incomplete observation.

A KNN Imputer with **5 nearest neighbors** was evaluated.

Since KNN uses distance calculations, numerical variables were standardized before imputation.

KNN Results

Column	MAE	RMSE
Age	10.935	13.426
Positive Feedback Count	5.604	9.145

The results showed that KNN was able to estimate the masked values, but the errors were relatively high compared with the other evaluated methods.

9. MICE Imputation

MICE stands for **Multiple Imputation by Chained Equations**.

MICE estimates missing values iteratively by using relationships between multiple variables. In this project, a MICE-style approach was implemented using `IterativeImputer`.

MICE Results

Column	MAE	RMSE
Age	10.003	12.323
Positive Feedback Count	3.024	5.777

MICE produced lower MAE and RMSE values than KNN for both evaluated numerical attributes.

10. Predictive Modeling

The third approach used machine learning to predict missing values.

A **Random Forest Regressor** was used for numerical predictive modeling.

For example, when predicting `Age`, the model used other available numerical variables such as:

- Rating
- Recommended IND
- Positive Feedback Count

Similarly, other available features were used to predict `Positive Feedback Count`.

Predictive Modeling Results

Column	MAE	RMSE
Age	10.034	12.381
Positive Feedback Count	3.050	5.796

Predictive modeling performed considerably better than KNN and was very close to MICE.

11. Comparison of Imputation Techniques

The three evaluated techniques were compared using MAE and RMSE.

Age

Method	MAE	RMSE
KNN	10.935	13.426
MICE	10.003	12.323
Predictive Modeling	10.034	12.381

Positive Feedback Count

Method	MAE	RMSE
KNN	5.604	9.145
MICE	3.024	5.777
Predictive Modeling	3.050	5.796

Lower MAE and RMSE indicate better predictions.

MICE achieved the lowest error for both tested numerical attributes.

Therefore, **MICE was selected as the best-performing numerical imputation method in this experiment.**

12. Handling Actual Missing Values

The actual missing values in the dataset were mainly present in text and categorical attributes.

It would not be appropriate to blindly apply a numerical imputation technique to these variables.

Therefore, a column-specific strategy was used.

Title

The `Title` column contained 3,810 missing values.

Missing titles were replaced with:

"No Title"

An additional `Title_Missing_Flag` variable was created to preserve information about whether the title was originally missing.

Review Text

The `Review Text` column contained 845 missing values.

Missing reviews were replaced with:

"No Review"

A `Review_Text_Missing_Flag` variable was also created.

Artificial customer reviews were not generated because doing so could introduce information that the customers never actually provided.

Division Name, Department Name and Class Name

These three categorical attributes each contained 14 missing values.

`Clothing ID` was used to identify other records belonging to the same product. The most common known category for the corresponding Clothing ID was used to fill the missing category where available.

If a product-level category was unavailable, the overall mode was used as a fallback.

This approach used the structural relationship between the product identifier and its category rather than assigning an arbitrary value.

13. Manual Selection of Methods

The final methods were selected according to the data type, statistical evaluation, and structural characteristics of each variable.

Column	Type	Selected Treatment
Age	Numerical	MICE — experimental best
Positive Feedback Count	Numerical	MICE — experimental best
Title	Text	Missing label + indicator
Review Text	Text	Missing label + indicator
Division Name	Categorical	Clothing ID-based imputation
Department Name	Categorical	Clothing ID-based imputation
Class Name	Categorical	Clothing ID-based imputation

This demonstrates why a single imputation method should not automatically be applied to every variable.

14. Before and After Analysis

Before preprocessing, the dataset contained **4,697 missing entries**.

After the selected treatments were applied, the missing values were eliminated from the cleaned dataset.

The numerical attributes did not originally contain missing values, so their real observed values were left unchanged during final cleaning.

This helped preserve the original numerical statistics of the dataset.

The categorical distributions were expected to change only minimally because only 14 records were missing from each of the affected categorical attributes.

15. Key Findings

The analysis produced several important findings:

1. **Title** contained the highest amount of missing data at approximately 16.22%.
2. **Review Text** contained approximately 3.60% missing data.
3. The categorical variables had only about 0.06% missing data each.
4. MICE produced lower MAE and RMSE than KNN and Random Forest predictive modeling in the controlled numerical experiment.
5. Predictive modeling performed very close to MICE.
6. KNN produced the highest errors among the three tested numerical methods.
7. Text attributes required a different strategy because numerical imputation techniques are not directly appropriate for natural-language strings.

8. Product-level information from `Clothing ID` provided a meaningful way to handle missing product categories.
 9. The final preprocessing strategy was selected separately for each type of variable.
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16. Conclusion

This project successfully analyzed and handled missing values in the **Women's Clothing: Trends and Analysis in the Fashion Industry** dataset.

Three major numerical imputation techniques—**KNN, MICE, and predictive modeling using Random Forest**—were evaluated through a controlled masking experiment. Their performance was compared using MAE and RMSE.

Among the tested techniques, **MICE achieved the lowest MAE and RMSE for both Age and Positive Feedback Count**, making it the best-performing method in the numerical experiment.

However, the actual missing values were mainly found in text and categorical variables. Therefore, instead of applying one technique blindly to every column, the final preprocessing strategy was selected according to the nature of each attribute.

Text fields were handled using explicit missing labels and missing indicators, while categorical product information was recovered using known information from records sharing the same Clothing ID, with mode as a fallback.

The final result was a cleaned dataset suitable for further exploratory analysis, visualization, fashion-trend analysis, and machine-learning applications.

17. Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Jupyter Notebook

Machine Learning / Imputation Techniques

- KNN Imputer
- MICE / Iterative Imputer
- Random Forest Regressor

Evaluation Metrics

- Mean Absolute Error (MAE)
 - Root Mean Squared Error (RMSE)
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18. Final Outcome

Original Missing Entries: 4,697

Final Missing Entries: 0

Best Numerical Method: MICE

Final Output: Cleaned Women's Clothing Dataset

The preprocessing stage of the project was successfully completed and the resulting dataset is ready for further fashion-industry analysis.